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Background and objective

Light microscopy remains central to malaria diagnosis in endemic settings [1]. Deep models trained on curated thin-smear data may not retain performance when staining, image scale, or smear preparation changes [2,3].

MORPHA encodes stage-conditional morphology from BBBC041 as a soft training penalty [4]. We evaluate transfer from thin-smear sources to Lacuna African field microscopy [5], including the boundary between thin-smear training and thick-smear deployment.

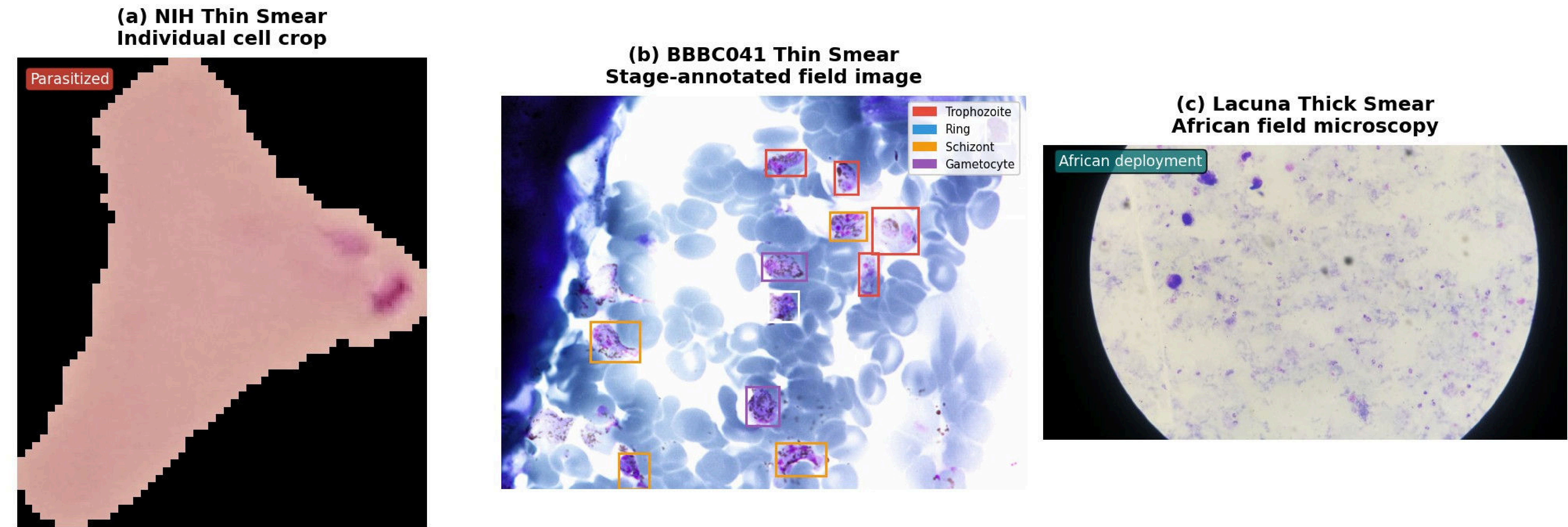


Figure 1. Source and target domains: NIH thin-smear cell crops [2], BBBC041 stage-annotated thin-smear images [4], and Lacuna African field microscopy [5].

Methods

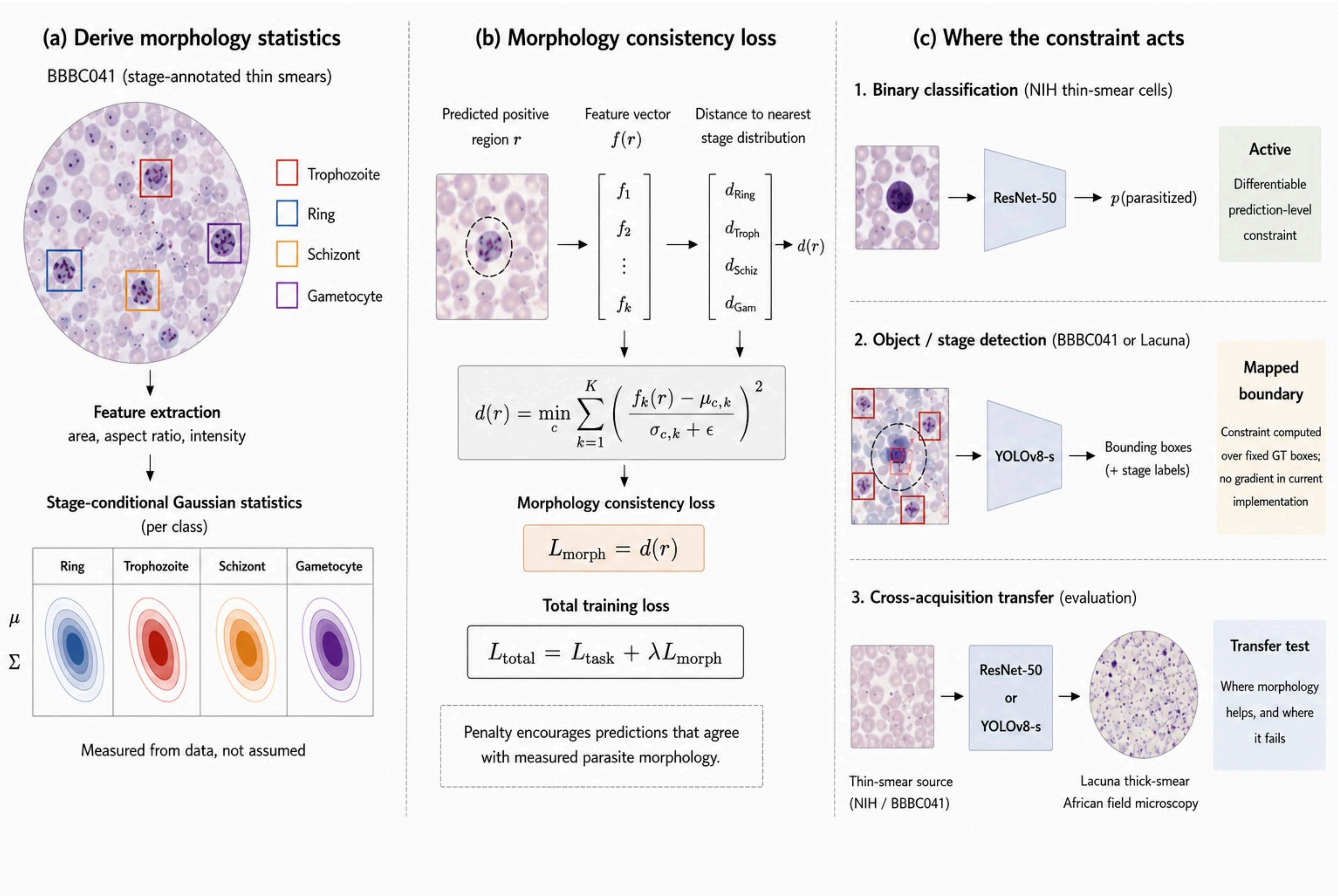


Figure 2. MORPHA derives stage statistics, applies the available constraint within each annotation regime, and evaluates cross-acquisition transfer.

Results

Cross-dataset generalisation					
Model	Source	Target	Within	Cross	Drop
Binary-UC	NIH	Lacuna	0.9739	0.5783	0.3956
Binary-MC	NIH	Lacuna	0.9729	0.6992	0.2737
Stage-UC	BBBC041	Lacuna	0.832±	0.000	0.832
Stage-MC†	BBBC041	Lacuna	0.832±	0.000†	0.832†

Drop = within − cross. ± BBBC041 trophozoite mAP50. † Predicted, not measured.

Conclusions

- MORPHA reduced the binary cross-dataset generalisation drop by 30.8% at negligible within-domain cost.
- Measured morphology contributed beyond the random-statistics control, although label smoothing and a confidence penalty achieved stronger raw transfer.
- Thin-smear detection did not transfer to thick-smear field images. Substantial out-of-distribution overconfidence remained.

Formulation and training

$$d(r) = \min_c \sum_k \left(\frac{f_k(r) - \mu_{c,k}}{\sigma_{c,k} + \epsilon} \right)^2$$

$$L_{\text{morph}}^{\text{bin}} = \frac{1}{|B|} \sum_r p_1(r) d(r)$$

$$L_{\text{total}} = L_{\text{task}} + \lambda L_{\text{morph}}, \quad \lambda = 0.1$$

Stage statistics. Area, aspect ratio, and intensity were estimated from 79,672 BBBC041 instances [4].

Binary model. An ImageNet-pretrained ResNet-50 [6] was trained for 50 epochs with AdamW. The selected constraint weight was $\lambda = 0.1$.

Detection model. YOLOv8-s was used for object and stage detection. The implemented constraint operates over fixed ground-truth boxes and therefore contributes no gradient.

External evaluation. Transfer was assessed on 2,432 matched Lacuna images [5]. Calibration and predictive uncertainty were estimated with Monte Carlo dropout [7].

Content and regulariser controls					
Model	NIH F1	NIH ECE	Lac F1	Lac ECE	Reduction
Binary-UC	0.9739	0.0165	0.5783	0.4827	—
Binary-MC	0.9729	0.0084	0.6992	0.3517	30.8%
MC-Random	0.9768	0.0071	0.6813	0.2974	25.3%
Label smooth.	0.9692	0.0367	0.7570	0.2001	46.4%
Conf. penalty	0.9741	0.0031	0.7404	0.2530	40.9%

Lacuna matched set ($n = 2,432$). Reduction is relative to Binary-UC. Single-pass ECE.

References

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